State-Dependent Macroeconomic Forecasting for Equity Returns

Stanford CS229 Project

Jagat Brahma
Department of Computer Science
Stanford University
jbrahma@stanford.edu

Abstract

Predicting the stock market from economic data is notoriously difficult - the signal is weak and easily drowned out by noise. This project builds a multi-stage machine learning pipeline called SSRF (State-Dependent Supervised Screening & Regularized Factor model) that forecasts S&P 500 returns from over 100 macroeconomic indicators. We evaluate at two horizons. On monthly 1-month-ahead returns, the pipeline performs near chance (47–57% directional accuracy), confirming the extreme difficulty of short-term prediction. On 3-month overlapping forward returns, the best models achieve up to 74.5% directional accuracy, exceeding the 67% overlapping always-long baseline. The top configuration achieves a Sharpe ratio of 0.87 out-of-sample from 2000–2026. An asymmetric leverage strategy (2.5$\times$ long / 0.25$\times$ short) boosts Sharpe for long-biased models but has mixed effects on the strongest predictors (CatBoost, Ensemble).

1 Introduction

Can economic data predict the stock market? This question has fascinated researchers for decades (Goyal and Welch, 2008). The challenge is that financial markets are noisy and driven by factors far beyond government-reported statistics. Yet intuitively, when the economy is strong (employment is high, factories are busy, consumers are spending), corporate profits-and therefore stock prices-should rise.

The input to our system is a set of roughly 130 monthly economic statistics published by the Federal Reserve and other agencies: employment, industrial production, housing starts, interest rates, consumer sentiment, and inflation. The output is a prediction of whether the S&P 500 index (the most widely- followed U.S. stock market benchmark) will go up or down, evaluated at both monthly and 3-month forward horizons. We designed a four-stage pipeline that filters out noisy features, extracts hidden economic factors, and adjusts predictions based on market conditions. We evaluate using "walk-forward" backtesting, simulating real-time performance over the last 26 years with no future data leakage.

2 Related Work

Goyal and Welch (2008) famously showed that most economic variables studied over 30 years could not reliably predict the stock market out-of-sample. Campbell and Thompson (2008) pushed back, arguing that even modest predictability is economically valuable and introducing the out-of-sample R^2_{OOS} metric. Huang et al. (2022) proposed "Scaled PCA," which weights features by predictive power before dimension reduction - a key inspiration. Zou and Hastie (2005) developed the Elastic Net, combining L1 (lasso) and L2 (ridge) penalties, well-suited for datasets with many correlated predictors. Our work differs by combining screening, scaling, factor extraction, and regime interaction into a single defensible pipeline, and by emphasizing statistical rigor (bootstrap confidence intervals, permutation tests, Diebold-Mariano tests) to avoid overclaiming on a noisy problem.

3 Dataset and Features

Economic indicators. Our primary data source is the FRED-MD database (McCracken and Ng, 2016), a collection of 134 monthly macroeconomic time series maintained by the Federal Reserve Bank of St. Louis ("FRED" = Federal Reserve Economic Data). The indicators span output (GDP, industrial production), labor (employment, housing starts), consumption (retail sales), money/credit (M2, bank credit), interest rates (Treasury yields, corporate spreads), prices (CPI, PPI), and sentiment (consumer confidence, VIX). Data covers January 1979 through June 2026. We fill missing values forward and cap extreme outliers at the 1st and 99th percentiles.

Target variable. We evaluate at two forecast horizons. The first target is the monthly percentage change of the S&P 500 index (Yahoo Finance, ticker `^GSPC`): $\frac{P_t - P_{t-1}}{P_{t-1}} \times 100$. Features are aligned so that economic data from month t predicts returns for month $t+1$, mimicking a realistic scenario where macro reports are released with a lag.

The second target is the 3-month overlapping forward return: $y_t = \frac{P_{t+3} - P_t}{P_t} \times 100$, evaluated at monthly frequency. This produces monthly predictions where each forecasts the cumulative return over the following quarter. Consecutive predictions share 2/3 of their observation window, which inflates apparent significance but preserves sample size. We report results for both targets and note that on 3-month returns the correct always-long baseline rises to 67% (the market rose in 67% of overlapping 3-month windows from 2000–2026).

Train/test splits. We use an expanding "walk-forward" design: at each step the model trains on all data up to the current month and predicts the next month. The first 120 months form the initial training set. For a cleaner evaluation, we also fix a split at January 2000 - the model trains only on pre-2000 data and is evaluated strictly on 2000–2026 with no retraining ("true OOS").

4 Methods

The SSRF pipeline has four stages, each addressing a specific challenge in macroeconomic forecasting:

Stage 1: Group-wise Screening. We have 134 features but only 540 months of data. Features are grouped into economic categories (output, labor, inflation, etc.), and within each category we retain only those with a t-statistic magnitude above 0.75. This prevents any single category from dominating. About 40–60 features survive.

Stage 2: Predictive Scaling ("Scaled PCA"). Before extracting factors, each surviving feature is scaled by its univariate regression slope against returns. This step, inspired by Huang et al. (2022), amplifies predictive features and suppresses noise.

Stage 3: Factor Extraction (PCA). Even after screening, the remaining features are highly correlated (they all track the same economy). Principal Component Analysis (PCA) extracts 10 summary factors capturing the main independent sources of variation. The first factor typically tracks the business cycle; later factors capture interest-rate or inflation dynamics.

Stage 4: Regime Interaction. Markets behave differently in calm vs. turbulent periods. We measure the current "regime" using rolling 12-month volatility - essentially asking "is this market calmer or more volatile than usual?" Each factor is multiplied by this regime measure, creating interaction terms that let the model learn different behavior under different conditions. We also test additional regime indicator variables (using Hidden Markov Model and volatility-based detectors) as supplementary features.

Final Regression. We test seven regressors: Elastic Net (Zou and Hastie, 2005) (L1+L2, $\alpha = 0.001$, ℓ_1-ratio$= 0.5$), unregularized linear regression, XGBoost (gradient-boosted trees), Random Forest, CatBoost, MLP (neural network), and an ensemble averaging two models (Linear + XGBoost). The SSRF pipeline refers to the four-stage feature engineering process; any regressor can be plugged in at Stage 4. Training is walk-forward: the window expands with each month.

Baselines. We compare against historical average (always predict the training mean), momentum (predict the same direction as last month), and random guessing.

Asymmetric leverage. We also test asymmetric position sizing: 2.5$\times$ long when predicting positive returns but only 0.25$\times$ short when predicting negative returns. This exploits the empirical observation that long signals are more reliable than short signals in macro forecasting.

5 Experiments / Results / Discussion

Evaluation metrics. We use four primary metrics:

- **Hit ratio (direction accuracy):** percentage of months where the predicted sign matches the actual sign (random = 50%).
- **Sharpe ratio:** annualized mean return divided by annualized volatility, measuring risk-adjusted return.
- R^2_{OOS}**:** compares the model's squared errors against the historical average (positive = beats the average).
- **Bootstrap 95% confidence interval for Sharpe:** computed by resampling the return stream 1000 times (excludes zero if significant).

Main results. Table 1 shows the primary out-of-sample results (2000–2026).

Table 1 reveals a striking divergence between the two horizons. On monthly 1-month-ahead returns (Panel A), all seven model types perform at or near chance levels (47–57% hit ratio). The always-long baseline of 62.5% beats every model, and all R^2_{OOS} values are strongly negative (-1.3 to -1.6), indicating that models fail to predict either the direction or magnitude of next-month returns. Momentum achieves 62.7% direction accuracy-essentially matching the always-long baseline-with a Sharpe of 0.50. These results confirm the well-known difficulty of predicting equity returns at a monthly horizon.

On 3-month overlapping forward returns (Panel B), the picture changes. The best-performing model, CatBoost, achieves 74.5% directional accuracy with a Sharpe of 0.85. The ensemble (equal-weighted average of Linear and XGBoost) achieves 70.3% directional accuracy and the highest Sharpe (0.87). Even the simplest model–linear regression and ElasticNet-reach 65.5% and 59.3% respectively. Here the always-long baseline is 66.6% (the market rose in 67% of overlapping 3-month windows from 2000–2026), so CatBoost's 74.5% represents a 7.9 percentage point improvement and the ensemble's 70.3% a 3.7 point improvement. All R^2_{OOS} values remain negative, meaning the models cannot reliably predict *magnitudes*, but the direction signal on 3-month returns is economically meaningful.

Caveat on overlapping returns. The 3-month forward returns are overlapping: consecutive predictions share 2/3 of the same observation window. This inflates apparent persistence and can overstate Sharpe ratios and statistical significance. The direction accuracy metric is more robust to overlap than portfolio metrics, but readers should interpret the 3-month results with this caveat in mind. For a conservative baseline, the non-overlapping 3-month always-long accuracy (where each 3-month block is independent) is 63.2%.

Walk-forward configurations on 3-month returns with expanding window give 66% hit ratio (Sharpe 0.60); fixed 60-month window gives 63% (Sharpe 0.53).

Asymmetric leverage. Table 2 shows the effect of 2.5$\times$ long / 0.25$\times$ short leverage with enhanced features including the CAPE ratio and put/call ratio:

Table 1: Out-of-sample results (2000–2026) at two horizons.

Panel A: Monthly Returns (1-month ahead)				
Model	Hit Ratio	Sharpe	R^2_{OOS}	Total Return
ElasticNet	55.0%	−0.00	−1.31	−4%
Linear	50.7%	−0.28	−1.41	−36%
XGBoost	56.7%	−0.00	−1.29	−6%
Random Forest	54.6%	0.05	−1.36	+5%
CatBoost	56.9%	0.03	−1.29	+2%
MLP	47.2%	−0.32	−1.57	−38%
Ensemble	52.8%	−0.22	−1.37	−34%
Momentum	62.7%	0.50	−0.003	+30%
Always Long	62.5%	—	0.000	—
S&P 500 B&H	—	0.66	—	+444%
Panel B: 3-Month Forward Returns (overlapping, monthly freq)				
Model	Hit Ratio	Sharpe	R^2_{OOS}	Total Return
ElasticNet	59.3%	0.33	−0.91	+82%
Linear	65.5%	0.74	−0.48	+357%
XGBoost	66.2%	0.82	−0.40	+395%
Random Forest	62.8%	0.61	−0.81	+286%
CatBoost	**74.5%**	0.85	−0.40	+641%
MLP	62.1%	0.33	−1.05	+61%
Ensemble	**70.3%**	**0.87**	−0.44	+486%
Always Long (overlap)	66.6%	—	0.000	—
Always Long (non-overlap)	63.2%	—	0.000	—
S&P 500 B&H	—	0.66	—	+444%

Table 2: Asymmetric leverage (2.5×/0.25×).

Model	Ann. Return	Ann. Vol.	Sharpe	Max. Drawdown
XGBoost	8.6%	10.4%	**0.84**	−9.1%
Ensemble	10.1%	12.8%	0.81	−11.5%
Linear	9.4%	13.1%	0.75	−9.4%
S&P 500 B&H	9.5%	13.9%	0.68	−46.7%

Asymmetric leverage has model-dependent effects. For the strongest models (Ensemble), Sharpe decreases slightly (0.87 to 0.81) at 2.5×/0.25×, while for weaker models (ElasticNet) Sharpe more than doubles (0.33 to 0.67). The primary benefit across all models is maximum drawdown reduction from equity-like levels (−47% B&H) to bond-like levels (−9% to −12%).

When applied to individual sectors, the model performs best on defensive sectors: Consumer Staples ($R^2_{OOS} = +0.04$, 64% hit ratio) and Healthcare ($R^2_{OOS} = +0.03$). It struggles with cyclical sectors like Energy ($R^2_{OOS} = -1.94$) and Financials ($R^2_{OOS} = -1.04$).

What didn't work. Several experiments failed or disappointed. Extreme leverage (25× long) destroyed every portfolio, with 72–84% drawdowns across all model types. The R^2_{OOS} remains negative in all configurations - the model cannot predict return magnitudes, only direction. Transaction costs, even at professional-tier rates (15 bps), consume 1–2% of annual returns, narrowing the advantage over buy-and-hold. These failures reinforce a core lesson: equity return prediction is a low signal-to-noise problem, and claimed successes should always be viewed skeptically without proper out-of-sample testing and statistical significance.

6 Reproducibility

The pipeline is available at `https://github.com/pellucide/sp500_macro_forecast-`. To reproduce the main 3-month result (ensemble, full 7-model sweep):

```
python run_all_models_oos.py --sweep --step-size 3
```

All CLI flags (model type, leverage, screening threshold, rebalance frequency, etc.) are documented in `docs/CLI_USAGE.md`. The pipeline provides a Python API and CLI for exploring beyond the experiments above.

7 Conclusion / Future Work

We built a multi-stage pipeline that achieves up to 74.5% directional accuracy on 3-month overlapping forward returns and a Sharpe of 0.87 out-of-sample from 2000–2026. On monthly 1-month-ahead returns, the pipeline performs near chance, confirming the extreme difficulty of short-term equity prediction. The signal in the macroeconomic data appears at the quarterly horizon rather than the monthly horizon. An improperly matched always-long baseline (monthly 62.5% vs. 3-month 66.6%) can create a misleading picture. Asymmetric leverage ($2.5\times/0.25\times$) reduces maximum drawdowns from equity-like (-47%) to bond-like (-9% to -12%) levels, though it slightly reduces Sharpe for the strongest predictors (Ensemble: 0.87 to 0.81) while more than doubling it for weaker models (ElasticNet: 0.33 to 0.67).

Three limitations suggest future work. First, we do not model the release lag of economic data - FRED statistics for a given month are often published weeks later. Second, the core regime proxy (rolling 12-month volatility percentile) could be replaced with a hidden Markov model for more accurate state estimation. Third, transaction costs were tested only in a limited analysis; a production system would need to account for slippage and market impact.

8 Contributions

Sole author. All code, analysis, and writing by Jagat Brahma.

References

John Y. Campbell and Samuel B. Thompson. Predicting excess stock returns out of sample: Can anything beat the historical average? *Review of Financial Studies*, 21(4):1509–1531, 2008.

Amit Goyal and Ivo Welch. A comprehensive look at the empirical performance of equity premium prediction. *Review of Financial Studies*, 21(4):1455–1508, 2008.

Dashan Huang, Fuwei Jiang, Guoshi Li, Guoshi Tong, and Guofu Zhou. Scaled PCA: A new approach to dimension reduction. *Management Science*, 68(3):1678–1701, 2022.

Michael W. McCracken and Serena Ng. FRED-MD: A monthly database for macroeconomic research. *Journal of Business & Economic Statistics*, 34(4):574–589, 2016.

Hui Zou and Trevor Hastie. Regularization and variable selection via the elastic net. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 67(2):301–320, 2005.